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Unit – 5 : Session – 9 : SLO – 1 | Applications of Graphs

■ I. Fill in the Blanks

- A GRAPH is a collection of vertices connected by edges.
- GPS navigation systems use DIJKSTRA'S algorithm to find the shortest path.
- A DIRECTED graph has edges with direction associated with them.
- KRUSKAL'S / PRIM'S algorithm is used to find a Minimum Spanning Tree in a graph.
- Social media platforms can be modeled using UNDIRECTED (friendship) or DIRECTED (follow) graphs.

■ II. True or False

- Dijkstra's algorithm works with graphs having negative edge weights — FALSE.
- In social networks, people can be represented as nodes — TRUE.
- A tree is a special case of a graph that contains cycles — FALSE (a tree is acyclic).
- Graphs are not useful in real-world applications — FALSE.
- Kruskal's and Prim's algorithms are used to find the shortest path in a graph — FALSE (they find the MST).

■ III. Match the Following

1. Web page linking — b) Directed Graph
2. Telephone network optimization — d) Kruskal's Algorithm
3. Flight route planning — c) Dijkstra's Algorithm
4. Cable TV wiring — a) Minimum Spanning Tree
5. GPS Navigation — e) Weighted Graph (with Dijkstra's shortest path)

■ IV. Short Answer Questions

1. Graph: a set of vertices connected by edges, $G = (V, E)$. Applications: social networks, road networks/GPS.
2. Google Maps/GPS: intersections and locations are vertices, roads are weighted edges (distance/time);
 - Dijkstra's (or A*) computes the shortest/fastest route between source and destination.
3. Directed vs undirected: directed edges are one-way ($u \rightarrow v$), undirected edges are mutual ($u - v$).
4. MST: a subset of edges connecting all vertices with minimum total weight and no cycles; used in network design — laying cables, pipelines, and electrical grids at minimum cost.
5. Computer networks: routers/computers are vertices, links are weighted edges (latency/bandwidth);
 - routing protocols use shortest-path algorithms; network design uses MSTs.